

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 4: DESCRIBING MOTION AROUND US [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH04/SET-B****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 A car covers the first half of its total journey distance at a speed of 40 km/h and the remaining half distance at 60 km/h. What is the average speed for the entire trip?

- (A) 50.0 km/h (B) 48.0 km/h
(C) 52.5 km/h (D) 45.0 km/h

Q.2 A particle moves along a straight line. Its displacement in the n^{th} second is given by $s_n = u + \frac{a}{2}(2n - 1)$. If a body starts from rest with $a = 4 \text{ m/s}^2$, what distance does it travel during the 5th second?

- (A) 20 m (B) 18 m
(C) 50 m (D) 10 m

Q.3 A bullet moving with a velocity of 20 m/s penetrates 5 cm into a wooden block before coming to rest. What is the uniform retardation produced by the block?

- (A) 4000 m/s² (B) 2000 m/s²
(C) 8000 m/s² (D) 400 m/s²

Q.4 The ratio of distances fallen by a freely dropped body under gravity in the 1st, 2nd, and 3rd seconds of its motion (Galileo's odd number rule) is:

- (A) 1 : 2 : 3 (B) 1 : 4 : 9
(C) 1 : 3 : 5 (D) 1 : 1 : 1

Q.5 A train 100 m long crosses a bridge 400 m long at a uniform speed of 72 km/h. What is the time taken to cross the bridge completely?

- (A) 20 s (B) 25 s
(C) 15 s (D) 30 s

Q.6 The velocity-time graph of an object shows a triangle of base 8 s and height 16 m/s. The total displacement of the object is:

- (A) 128 m (B) 64 m
(C) 32 m (D) 16 m

Q.7 An athlete completes one round of a circular track of diameter 200 m in 40 s. What will be the displacement at the end of 2 minutes and 20 seconds?

- (A) Zero (B) 200 m
(C) 2200 m (D) 100 m

Q.8 A driver travelling at 25 m/s sees an obstruction and takes 0.4 s (reaction time) before hitting the brakes. The brakes produce a deceleration of 5 m/s². The total stopping distance is:

- (A) 62.5 m (B) 72.5 m
(C) 52.5 m (D) 82.5 m

Q.9 A stone is thrown vertically upwards with velocity u and returns to the thrower's hand in total time T . What is the initial velocity u in terms of g and T ?

- (A) $u = gT$ (B) $u = \frac{gT}{2}$
(C) $u = 2gT$ (D) $u = \frac{gT^2}{2}$

Q.10 If the displacement of an object is proportional to the square of time ($s \propto t^2$), the object is moving with:

- (A) Uniform velocity (B) Uniform non-zero acceleration
(C) Increasing acceleration (D) Decreasing speed

Q.11 A body starts from rest and covers a distance s_1 in the first 10 s and an additional distance s_2 in the next 10 s under uniform acceleration. The relation between s_1 and s_2 is:

- (A) $s_2 = s_1$ (B) $s_2 = 2s_1$
(C) $s_2 = 3s_1$ (D) $s_2 = 4s_1$

Q.12 The slope of a velocity-time graph of a moving body is negative. This indicates that:

- (A) The body is speeding up (B) The body is accelerating uniformly in direction of motion
(C) The body is undergoing retardation (deceleration) (D) The body has stopped moving

Q.13 What is the angle between the velocity vector and the acceleration vector of a body undergoing uniform circular motion?

- (A) 0° (B) 45°
(C) 90° (D) 180°

Q.14 A body moves with initial velocity 10 m/s and decelerates at 2 m/s². What is its velocity after covering a distance of 21 m?

- (A) 4 m/s (B) 8 m/s
(C) 6 m/s (D) 2 m/s

Q.15 A particle covers equal distances in equal intervals of time along a circular path. Its velocity is:

- (A) Constant in magnitude and direction (B) Constant in magnitude but variable in direction
(C) Zero (D) Decreasing continuously

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): A body can have zero velocity and non-zero acceleration simultaneously.

Reason (R): At the highest point of vertical projectile motion, the instantaneous velocity is zero while acceleration due to gravity is 9.8 m/s^2 downwards.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): The distance travelled by a moving object can never be less than the magnitude of its displacement.

Reason (R): Distance is the actual scalar path length, whereas displacement is the straight-line shortest vector between start and finish.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Uniform circular motion is an accelerated motion.

Reason (R): An object in uniform circular motion moves with variable speed.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Two bodies of different masses dropped from the same height in a vacuum reach the ground at the exact same instant.

Reason (R): Acceleration due to gravity is independent of the mass of the falling object.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): If the velocity-time graph of a body is a curve, its acceleration is variable.

Reason (R): The slope of a velocity-time graph represents the acceleration of the body.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: High-Speed Train Kinematics Profile

A high-speed electric bullet train starts from rest at Station X ($t = 0$) and accelerates uniformly at 1.5 m/s^2 for 20 seconds. It then travels at constant cruise speed for 60 seconds. Finally, entering Station Y, the driver applies regenerative brakes producing a uniform deceleration of 3.0 m/s^2 until the train comes to a complete halt.

Q.21 What is the maximum cruise speed reached by the train at the end of 20 seconds?

- (A) 15 m/s (B) 30 m/s
(C) 45 m/s (D) 20 m/s

Q.22 What distance does the train cover during the initial 20 seconds of acceleration?

- (A) 150 m (B) 300 m
(C) 600 m (D) 450 m

Q.23 What distance does the train cover during the 60 seconds of uniform cruise speed?

- (A) 1800 m (B) 1200 m
(C) 900 m (D) 2400 m

Q.24 How long does the braking phase take to bring the train to a complete stop?

- (A) 5 s (B) 10 s
(C) 15 s (D) 20 s

Q.25 What is the total track distance between Station X and Station Y?

- (A) 2100 m (B) 2250 m
(C) 2400 m (D) 1950 m

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.1	(B)	When two equal distances are covered at speeds v_1 and v_2 , average speed is the harmonic mean: $v_{avg} = \frac{2 v_1 v_2}{v_1 + v_2} = \frac{2(40)(60)}{40 + 60} = \frac{4800}{100} = 48.0 \text{ km/h.}$
Q.2	(B)	$s_5 = 0 + \frac{4}{2}(2 \times 5 - 1) = 2(10 - 1) = 2 \times 9 = 18 \text{ m.}$
Q.3	(A)	$u = 20 \text{ m/s}$, $v = 0$, $s = 0.05 \text{ m}$. $v^2 = u^2 - 2as \Rightarrow 0 = 400 - 2a(0.05) \Rightarrow 0.1a = 400 \Rightarrow a = 4000 \text{ m/s}^2$.
Q.4	(C)	$s_n \propto (2n - 1)$. For $n=1, 2, 3$, the ratios are $(2(1)-1) : (2(2)-1) : (2(3)-1) = 1 : 3 : 5$.
Q.5	(B)	$\text{Total distance} = 100 + 400 = 500 \text{ m}$. $\text{Speed} = 72 \times \frac{5}{18} = 20 \text{ m/s}$. $\text{Time} = \frac{500}{20} = 25 \text{ s}$.
Q.6	(B)	$\text{Displacement} = \text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 16 = 64 \text{ m}$.
Q.7	(B)	Total time $t = 140 \text{ s}$. Number of rounds $n = 140 / 40 = 3.5$ rounds. After 3.5 rounds, the athlete is at the diametrically opposite point, so displacement $= \text{diameter} = 200 \text{ m}$.
Q.8	(B)	Thinking distance $= 25 \times 0.4 = 10 \text{ m}$. Braking distance $= \frac{v^2}{2a} = \frac{25^2}{2(5)} = \frac{625}{10} = 62.5 \text{ m}$. $\text{Total} = 10 + 62.5 = 72.5 \text{ m}$.
Q.9	(B)	Time of ascent $t = T/2$. At top, $v = 0 \Rightarrow 0 = u - g(T/2) \Rightarrow u = \frac{gT}{2}$.
Q.10	(B)	From $s = \frac{1}{2}at^2$, if a is constant, $s \propto t^2$. Hence, uniform non-zero acceleration.
Q.11	(C)	$s_1 = \frac{1}{2}a(10)^2 = 50a$. Total distance in 20 s: $s_{total} = \frac{1}{2}a(20)^2 = 200a$. Distance $s_2 = 200a - 50a = 150a$. Hence $s_2 = 3s_1$.
Q.12	(C)	A negative slope on a velocity-time graph indicates negative acceleration (retardation/braking).
Q.13	(C)	In uniform circular motion, velocity is tangential while centripetal acceleration is radially inward. They are strictly mutually perpendicular (90°).
Q.14	(B)	$v^2 = u^2 - 2as = 10^2 - 2(2)(21) = 100 - 84 = 16 \Rightarrow v = 4 \text{ m/s}$... Wait! $\sqrt{16} = 4 \text{ m/s}$! Let's check options: A is 4 m/s.
Q.15	(B)	Speed is constant, but velocity continuously changes direction tangent to the circle.
Q.16	(A)	At maximum height, an upward-projected object momentarily stops ($v = 0$), but gravitational force still acts, giving non-zero acceleration (g downwards).
Q.17	(A)	Because a straight line is the shortest distance between two points, $ \text{Distance} \geq \text{Displacement} $ always.
Q.18	(C)	Assertion is true (accelerated because direction changes). Reason is false because speed is constant in *uniform* circular motion.
Q.19	(A)	In vacuum with no air resistance, $g = GM/R^2$ is identical for all masses, so both take $t = \sqrt{2h/g}$ to hit the ground.
Q.20	(A)	Because slope equals acceleration, a non-linear curved graph has changing slope at different tangents, meaning variable acceleration.
Q.21	(B)	$v = u + at = 0 + 1.5 \times 20 = 30 \text{ m/s}$ (108 km/h).
Q.22	(B)	$s_1 = \frac{1}{2}at^2 = \frac{1}{2}(1.5)(20^2) = \frac{1}{2} \times 1.5 \times 400 = 300 \text{ m}$.
Q.23	(A)	$s_2 = v \times t = 30 \text{ m/s} \times 60 \text{ s} = 1800 \text{ m}$.
Q.24	(B)	$t_{brake} = \frac{v - 0}{a} = \frac{30}{3.0} = 10 \text{ s}$.

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.25	(B)	$s_3 = \frac{v^2}{2a} = \frac{30^2}{2(3)} = \frac{900}{6} = 150$ m. Total distance $s = 300 + 1800 + 150 = 2250$ m (2.25 km).